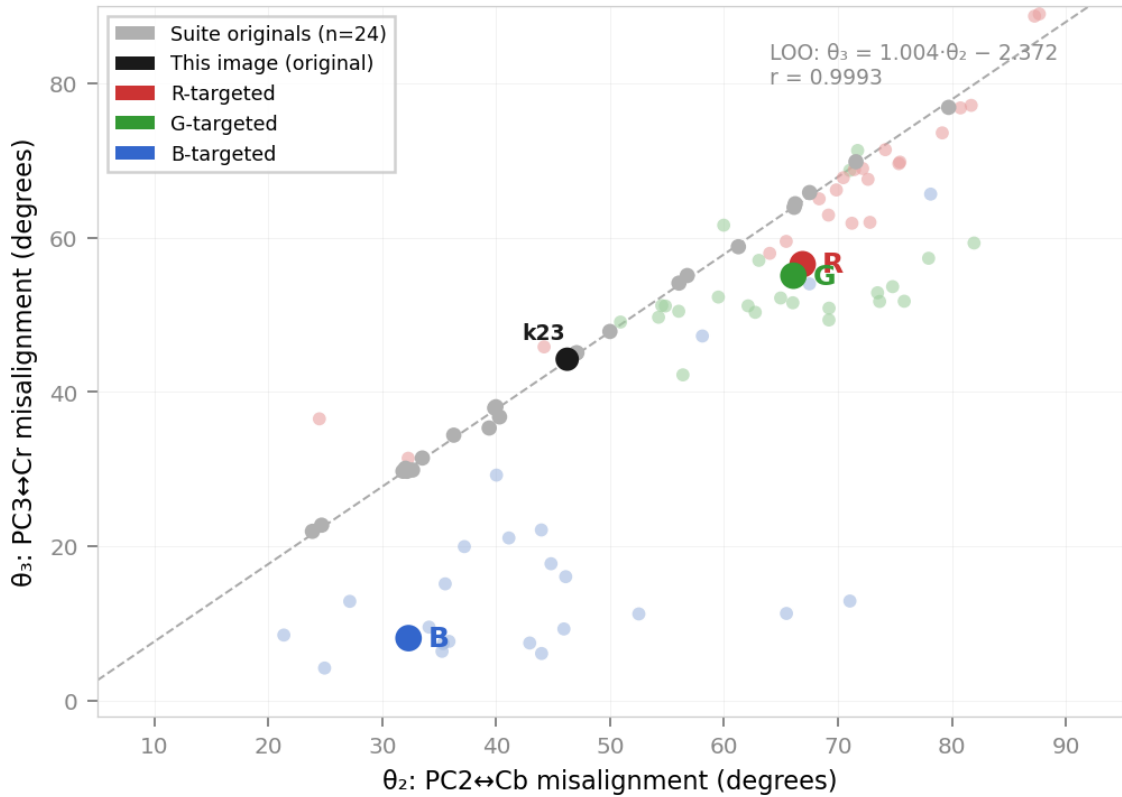


Compression Profile: KODIM23 768x512 | 24-bit RGB | PNG (lossless)

two macaws | Three-Dimensional (PC1 = 70.3%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.6147	0.4231	0.6408	0.559	46.23°	44.26°	+0.22°
R-targeted	-0.0398	-0.2164	0.9061	0.387	66.9°	56.58°	-8.22°
G-targeted	0.0703	0.9091	0.1650	0.382	66.09°	55.09°	-8.89°
B-targeted	0.9380	0.1967	0.1749	0.436	32.3°	8.09°	-21.97°

R-targeted isolates red (R-G, R-B < 0.22) while preserving G-B coupling at 0.906.
G-targeted isolates green (R-G, G-B < 0.17) while preserving R-B at 0.909.
B-targeted preserves R-G at 0.938 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold ($r = 0.9993$).

Compression Profile: KODIM23

RGB | PNG (lossless)

two macaws | Three-Dimensional (PC1 = 70.3%)

768x512 | 24-bit

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

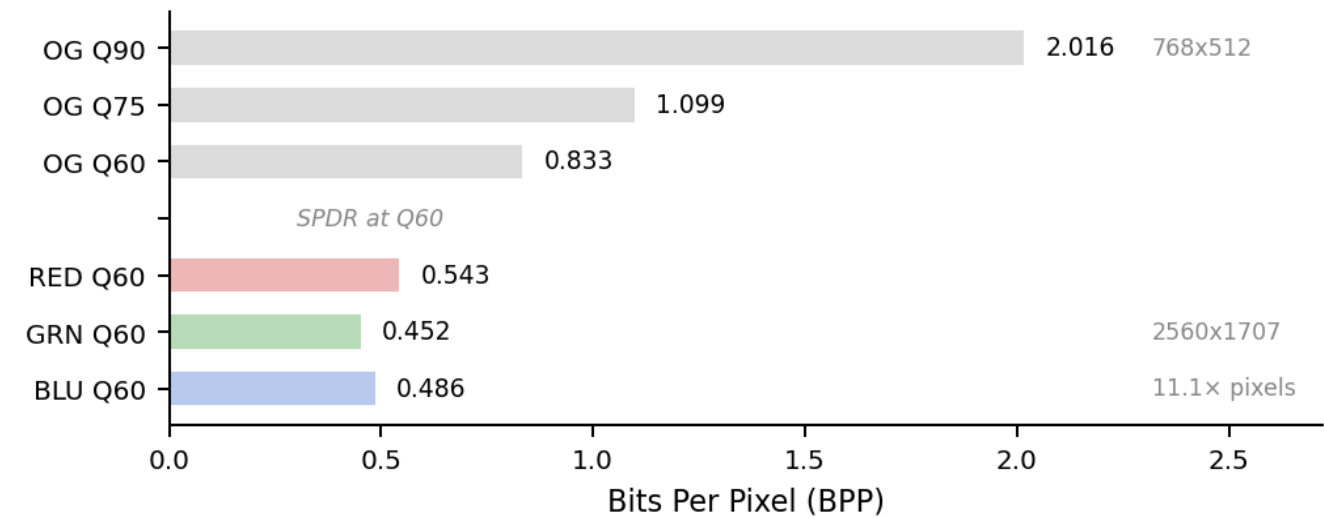
ORIGINAL (kodim23)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	768x512	544.5 KB	11.344	393,216
JPEG Q90	JPEG	768x512	96.8 KB	2.016	393,216
JPEG Q75	JPEG	768x512	52.8 KB	1.099	393,216
JPEG Q60	JPEG	768x512	40.0 KB	0.833	393,216

REDISTRIBUTED (kodim23 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	2560x1707	289.5 KB	0.543	4,369,920
GRN Q60	JPEG	2560x1707	241.2 KB	0.452	4,369,920
BLU Q60	JPEG	2560x1707	259.4 KB	0.486	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1x the original pixel count.

Compression Profile: KODIM23 768x512 | 24-bit RGB | PNG (lossless)

two macaws | Three-Dimensional (PC1 = 70.3%)

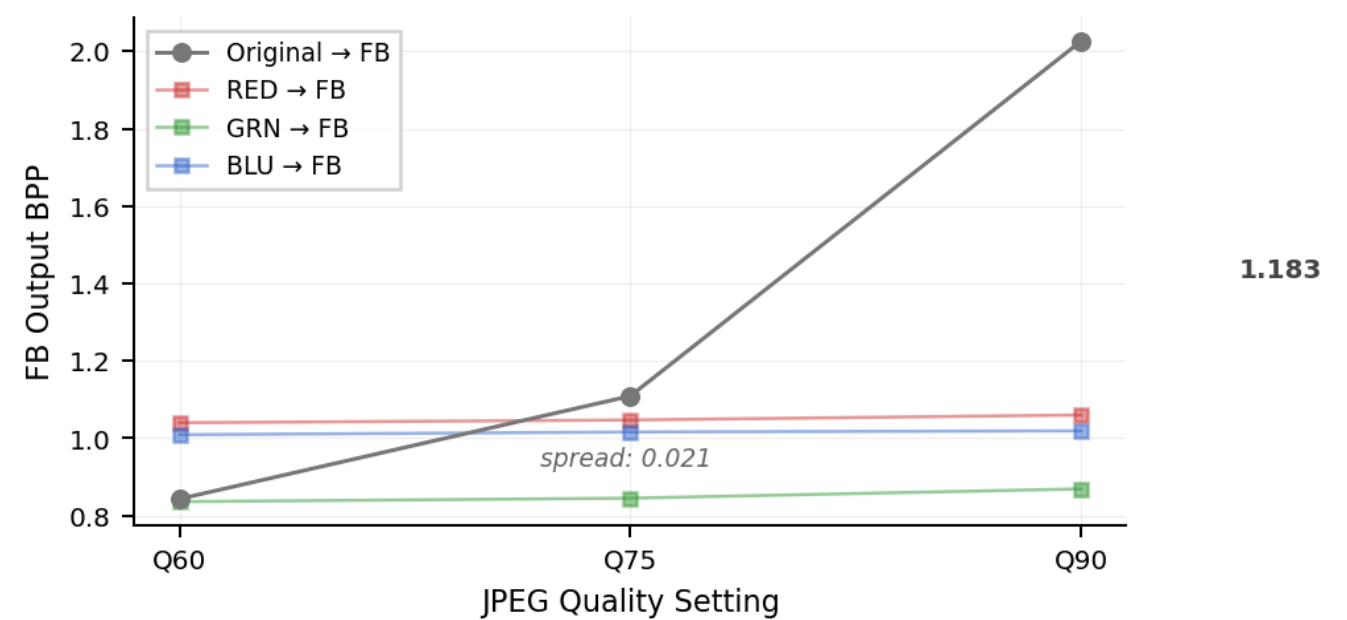
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	768x512	768x512	1.766	—
R-targeted → FB	2560x1707	2048x1366	1.060	40.0%
G-targeted → FB	2560x1707	2048x1366	0.869	50.8%
B-targeted → FB	2560x1707	2048x1366	1.019	42.3%
ABC (no perturbation) → FB	2560x1707	2048x1366	1.027	41.8%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.5589 vs 0.559 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.183 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.021 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim23

Best BPP (Q60, direct)	0.452	2560x1707, 4,369,920 px
Best BPP through FB	0.869	2048x1366 output
vs OG Q90 → FB	57.1%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216